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STABILIZING RLHF THROUGH ADVANTAGE MODEL AND SELECTIVE REHEARSAL

Baolin Peng[*], **Linfeng Song**[*], **Ye Tian, Lifeng Jin, Haitao Mi, Dong Yu**
Tencent AI Lab
{baolinpeng,lfsong,yaptian,lifengjin,haitaomi}@global.tencent.com

ABSTRACT

Large Language Models (LLMs) have revolutionized natural language processing, yet aligning these models with human values and preferences using RLHF remains a significant challenge. This challenge is characterized by various instabilities, such as reward hacking and catastrophic forgetting. In this technical report, we propose two innovations to stabilize RLHF training: (*i*) *Advantage Model*, which directly models advantage score *i.e.,* extra reward compared to the expected rewards and regulates score distributions across tasks to prevent reward hacking. (*ii*) *Selective Rehearsal*, which mitigates catastrophic forgetting by strategically selecting data for PPO training and knowledge rehearsing. Our experimental analysis on public and proprietary datasets reveals that the proposed methods not only increase stability in RLHF training but also achieve higher reward scores and win rates[1].

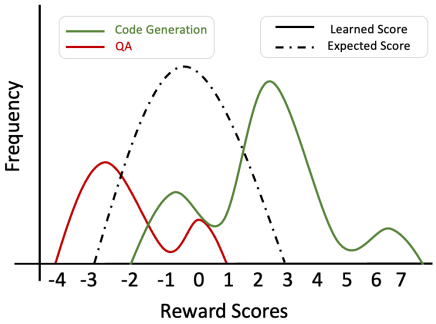
1 INTRODUCTION

Large language models (LLMs) have become a fundamental element in advancing natural language processing (NLP) and artificial intelligence (AI), showcasing an impressive ability to generate text that is both semantically and contextually relevant (OpenAI, 2023; Köpf et al., 2023; Touvron et al., 2023). Despite these advancements, LLMs have the risk of engaging in undesirable behaviors, such as fabricating information or producing biased, toxic, or even dangerous content, since LLMs are trained on a wide array of data, which can include low-quality sources. This has highlighted the necessities of LLM Alignments with human values, intentions, and preferences (Brown et al., 2020; Ouyang et al., 2022; Bai et al., 2022a; Glaese et al., 2022).

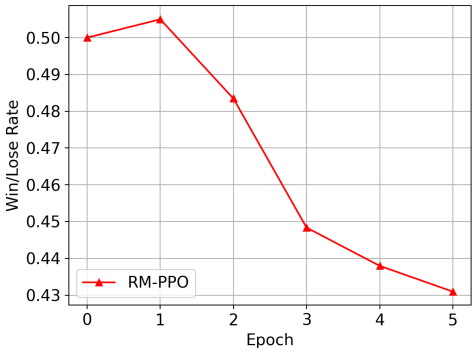
Many approaches have been put forth to address the challenge LLM Alignments (Bai et al., 2022a; OpenAI, 2023; Askell et al., 2021). Among these approaches, Reinforcement Learning from Human Feedback (RLHF) has demonstrated its efficacy in aligning language models with human preferences. RLHF serves as a key component of training SoTA LLMs including exemplars such as OpenAI's GPT-4 (OpenAI, 2023), Anthropic's Claude (Bai et al., 2022a), Google's Sparrow (Glaese et al., 2022), Bard, and Meta's Llama 2-Chat (Touvron et al., 2023). RLHF elevates the capabilities of LLMs beyond the mere modeling of the distribution of their training data. It endows LLMs with the capacity to adapt their text generation distribution in a manner that are preferred by humans.

However, training LLMs using RLHF is undoubtedly challenging, which demands an accurate and reliable reward model that approximates human judges, and a robust PPO algorithm for sustained policy improvements. Even with meticulous configurations, *instabilities*, *e.g.,* gibberish responses (but high-reward) (Stiennon et al., 2020; Skalse et al., 2022), forgetting learned knowledge, are usually observed during training, which leads to recurring failures. These instabilities have several causes: (*i*) different reward score distributions are learned for various categories by the reward model, potentially leading to *reward hacking* issues (Skalse et al., 2022), a phenomenon where the model finds unintended ways to maximize the reward. As depicted in Figure 1a, the reward model learns noticeable disparity in reward score distributions for Code Generation and QA tasks, 2 out of

[*]Equal Contribution
[1]Work in progress



(a) Reward score distributions.



(b) Win rate over the SFT model on the *forget set* evaluated by GPT-4.

Figure 1: *Left*: The distribution of reward scores for both the QA and Code Generation tasks. There is a noticeable disparity in the learned reward score distributions between the two tasks, despite the expectation that the distributions should be similar. *Right*: The win/loss rate over the SFT model on the forget set exhibits a significant decline. This drop in the win rate can be attributed to reward hacking and the phenomenon of catastrophic forgetting.

61 tasks present in the preference data. Even with reward score normalizations, the fluctuating means and variances can induce unexpected model behaviors, such as transferring the response patterns of Code Generations to QA examples due to the higher reward scores. (*ii*) over-optimizing with PPO on examples that were well-aligned with humans in the Supervised Fine-Tuning (SFT) stage triggers *catastrophic forgetting* issues (McCloskey & Cohen, 1989; Gupta et al., 2023; Khetarpal et al., 2022). Models tend to overlook what was learned during the SFT stage, *i.e.,* PPO model underperforms the SFT model on expert-aligned examples [2], as shown in Figure 1b.

Accordingly, in this technical report, we introduce two techniques to enhance the stability and effectiveness of the training of RLHF. Firstly, we propose *Advantage Model* to balance the reward score distributions across various categories, thus averting the reward hacking dilemma that is often induced by noticeable differences score distributions. This is achieved by directly modeling the advantage score, *i.e.,* the extra reward one response can obtain compared with the expected reward, and regulating the advantage score distribution dynamically during training, ensuring that the variances and means are maintained within a reasonable range. Secondly, we introduce the *Selective Rehearsal* to alleviate the catastrophic forgetting issue. We posit that not all data should be optimized equally in PPO training. As such, we propose a robust and effective data selector that automatically identifies what examples could be utilized for PPO training and should be used to rehearsal knowledge accumulated in the SFT stage, preventing the depreciation of the model's performance on expert-aligned examples over time. Experiments on both public and proprietary data have demonstrated that our Advantage Model successfully balances reward score distributions across various examples while preserves ranking precision, and guide PPO training to achieve a higher reward score and win rate compared to the SFT model. Furthermore, Selective Rehearsal is able to avoid over-optimizing by selecting the most suitable examples for PPO training, thereby sustaining the performance on expert-aligned examples.

Our contributions are summarized as follows:

- We analyze and identify several causes of instability in RLHF training, namely, imbalanced learned reward score distributions and over-optimization of certain PPO training data, which lead to reward hacking and catastrophic forgetting issues.
- We introduce the *Advantage Model* to balance reward score distributions across various categories, and the *Selective Rehearsal* strategy to discern which examples should be used

[2]Expert-aligned Examples are data samples that meet the standards and criteria delineated by experts and closely align with human preferences. These examples are used for SFT model training and evaluation.

for PPO training and which should be reserved for rehearsing knowledge accrued in the SFT stage.

- Through extensive experiments on both public and proprietary datasets, we demonstrate that the *Advantage Model* and *Selective Rehearsal* are able to stabilize RLHF training, achieving higher reward scores and win rates.

2 PRELIMINARY

In recent machine learning research, RLHF (Ouyang et al., 2022; Bai et al., 2022a) has emerged as a pivotal strategy for aligning LLMs to human goals (e.g. being helpful and harmless). RLHF typically follows the SFT phase, where SFT aligns a LLM with human objectives using teacher forcing on (prompt, response) pairs. However, despite this alignment, the LLM may still struggle with generalization when faced with unseen tasks.

Learning a reward function from interaction between LLMs and humans and optimizing LLMs with the learned reward function using reinforcement learning has been shown as an effective approach to solving the LLM alignment problem. Leike et al. 2018; Stiennon et al. 2020; Ouyang et al. 2022 proposed a method involving reinforcement learning from human feedback, where RMs are trained on a dataset of comparisons between two model outputs generated from the same input. The goal is to assign higher rewards to outputs preferred by human labelers over others. Typically, this is achieved by adding a value head that outputs a scalar value on pre-trained transformer-baesd LMs with last umembedding layer removed. Specifically, the reward modeling loss is as follows:

$$\mathcal{L}_{\text{RM}} = -E_{(x,y_c,y_r)\sim D^{\texttt{RM}}}[\log(\sigma(r_\theta(x,y_c) - r_\theta(x,y_r)))] \tag{1}$$

where $r_\theta(x,y)$ denotes the reward score for prompt x and response y with parameters θ, y_c is the preferred response of the pair y_c and y_r, and $D^{\texttt{RM}}$ is the complete of comparison dataset.

In what follows, Proximal Policy Optimization (PPO) (Schulman et al., 2017) is commonly adopted as the reinforcement learning algorithm to optimize a policy due to its strengths in stability and simplicity. Particularly, the PPO objective for policy π on a prompt dataset D is defined as:

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_{x\sim D^{\texttt{PPO}}, y\sim\pi_\phi(x)}\left[r_\theta(x,y) - \beta\log\left(\pi_\phi(y|x)/\pi^{\texttt{init}}(y|x)\right)\right] \tag{2}$$

where $r_\theta(x,y)$ represents the reward score on the (prompt, response) pair of (x,y); $\pi^{\texttt{init}}$ indicates the policy before RLHF, and it is kept constant during RLHF training; β is the coefficient for the KL-divergence term.

Besides PPO, rejection sampling (Touvron et al., 2023) recently gains interests as a simple way for aligning LLMs. As an offline policy learning algorithm, it adopts an iterative process. For each iteration n, it first constructs a new dataset D_n by selecting (x,y) pairs from the main policy π_ϕ based on criteria $\mathcal{F}$:

$$D_n^{\texttt{PPO}} = \{(x,y)\cdot\mathcal{F}(x,y)|\text{ such that } x\sim D^{\texttt{PPO}}, y\sim\pi_\phi(x)\} \tag{3}$$

where a commonly used criteria $\mathcal{F} = \mathbb{1}_{r_\theta(x,y)\geq\tau}$ includes only the samples with RM scores exceed a certain threshold τ. The policy is then updated by teacher forcing on $D_n^{\texttt{PPO}}$:

$$\mathcal{L}_{\text{RS}} = \mathbb{E}_{(x,y)\sim D_n^{\texttt{PPO}}}\sum_{t=1}^{|y|}\pi_\phi(y_t|y_{<t},x) \tag{4}$$

3 APPROACH

3.1 FROM REWARD MODEL TO ADVANTAGE MODEL

The learning objective of equation 1 primarily allows models to distinguish between human-preferred responses and alternative options. It relies only on score differences to assess the likelihood of one response being superior to another. In such case, two different model responses that are both preferred by humans could have dramatically different values. In addition, interpreting the scalar values themselves can be challenging.

In light of these considerations, we introduce the Advantage Model (AM) for reward modeling. Analogous to the concept of the advantage function in reinforcement learning, the Advantage Model, denoted as $a(x, y)$, quantifies the additional reward that response y can achieve over the expected reward e for prompt x. This is formally defined as:

$$a_\theta(x, y) = r_\theta(x, y) - \mathbb{E}_{y \sim \pi'(x)}[\frac{\pi_\phi(y|x)}{\pi'(y|x)} r_\theta(x, y)] \tag{5}$$

Here, the notation $y \sim \pi'(x)$ signifies all possible responses generated by a policy $\pi'(x)$ when given the input prompt x. Since the comparison data is typically collected in many batches with different SFT or PPO models, we introduce $\frac{\pi^\phi(y|x)}{\pi'(y|x)}$, the importance weight term to negate the bias introduced by the policy distribution shift. Intuitively, the extra reward gains of good response y_c and the reward losses of bad response y_r should be bounded by a margin m. As such, the training objective of AM consists of two parts, *ranking loss* that aligns with the formulation in Equation 1, and *bounding loss* to ensure the well-calibrated bounding of AM scores. It is formally defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{AM}} = -E_{(x,y_c,y_r)\sim D^{\text{RM}}}[\log(\sigma(a_\theta(x, y_c) - a_\theta(x, y_r))) \\ + \log(\sigma(m(x) - a_\theta(x, y_c))) + \log(\sigma(m(x) + a_\theta(x, y_r)))] \end{aligned} \tag{6}$$

where $m(x)$[3] is the function that defines the permitted margin for prompt x. However, it is infeasible to list every potential response to calculate the expected reward. To address this, we propose parameterizing the expected reward of the current policy, denoted as:

$$e_\tau(x) = \mathbb{E}_{y \sim \pi_\phi(x)}[r_\theta(x, y)] \tag{7}$$

By integrating the term representing the importance weight, we can reformulate the equation as follows:

$$a_\theta(x, y) = r_\theta(x, y) - \frac{N-K}{N} e_\tau(x) - \sum_{k=1}^{K} \frac{1}{N} \frac{\pi^\phi(y|x)}{\pi'_k(y|x)} r_\theta(x, y) \tag{8}$$

where N serves as a hyperparameter that harmonizes the emphasis placed on the current policy model relative to alternate policy models. K specifies the number of alternate policy models utilized for comparison data collection. Additionally, $\pi'_k(y|x)$ indicates the probability derived from the kth policy model.

3.2 PPO WITH SELECTIVE REHEARSAL

In addition, we propose Selective Rehearsal to maintain the skills that are already acquired before RLHF. Selective rehearsal takes two major steps: representative example discovery and rehearsal training.

Representative example discovery Given the policy π_ϕ and PPO training prompts with policy outputs $D^{\texttt{PPO}} = [(x_1, y_1), (x_2, y_2) \ldots]$, our goal is to select high-quality (x, y) pairs from $D^{\texttt{PPO}}$ that cover as many skills (e.g., solving algebra problems and writing resume) as possible. In order to let selected (x, y) pairs represent as many skills as possible, we first adopt a clustering algorithm (e.g. KMeans or Gaussian mixture) to separate $D^{\texttt{PPO}}$ into c clusters. To assure the representativeness and quality of the selected data, we only keep certain (x, y) pairs within each cluster that satisfy certain criteria regarding aspects such as advantage (reward) model score, entropy (low entropy indicates high confidence), human satisfaction rate or response length (higher length may indicate redundancy).

Here we adopt the SimCSE (Gao et al., 2021) sentence embedding[4] to represent the query x for each (x, y) pair before running a KMeans algorithm on these embeddings to be grouped into c clusters. We briefly study the influence of cluster number c in Section 4.3. Within each cluster, here we simply choose the top-k (x, y) pairs with the highest advantage model score (Eq. 3.1). We leave other strategies (e.g. combining advantage score with entropy score) in future work.

[3]We think that $m(x)$ may have a connection with the complexity or difficulty involved in learning the reward function for prompts similar to x. However, this is speculative and requires further investigation. We leave this aspect as a topic for future study and exploration. Throughout our experiments, we set $m(x)$ as 2.5.

[4]`https://huggingface.co/princeton-nlp/sup-simcse-roberta-base`

One reason we select our rehearsal data from the PPO training data with each response y being generated from the initial policy model is to enable a more fair and nuanced comparison, as no additional information is introduced. In other scenarios, the rehearsal (x, y) pairs could come from other important data sources representing specific skills (e.g. math-problem solving) the main policy are not expected to forget.

Rehearsal training After obtaining the rehearsal (x, y) pairs of all clusters, we shuffle them together to form the rehearsal dataset D_R and compute NLL loss on D_R as a supplement to the standard PPO loss defined in Equation 2:

$$\mathcal{L}_{\text{PPO-SR}} = \mathcal{L}_{\text{PPO}} + \gamma \mathbb{E}_{(x,y)\sim D_R} \sum_{t=1}^{|y|} \pi_\phi(y_t|y_{<t}, x) \tag{9}$$

where the coefficient for the NLL loss γ is empirically set to 0.01.

Rehearsal training is similar with rejection sampling and reinforced self-training (Gulcehre et al., 2023) by using self-generated ys of high reward model score for supervised training. However, rehearsal training captures multi-dimensional important aspects (e.g., diversity), while rejection sampling and reinforced self-training only consider reward model score.

Alternatively, one can view selective rehearsal as a means of amplifying the weight of the KL-divergence term in PPO training (Eq. 2) for crucial instances and their related counterparts.

4 EXPERIMENTS

4.1 DATASETS AND MODELS

RM datasets We conducted experiments on both English and Chinese datasets. For the English experiments, we utilized the HH-RLFH dataset (Bai et al., 2022a; Ganguli et al., 2022), which comprises 118k helpful and 42k harmless examples for training, and 8.5k for testing. It is worth noting that many studies train different RMs separately for helpful and harmless examples to achieve better performance. However, in our experiments, we did not distinguish between helpful and harmless examples.

For the Chinese dataset, we collected comparison examples with quantities similar to those used in LLaMA 2 (Touvron et al., 2023). Our annotation procedure operates as follows: First, we ask annotators to generate prompts based on a task spectrum. Next, we sample five responses from the same SFT model using varied sampling hyper-parameters. Finally, we distribute these responses to five annotators for ranking based on provided criteria. Following Bai et al. (2022a), the annotation criteria focuses on helpfulness and harmless.

PPO dataset We sampled queries from two popular domain-general datasts, COIG[5] and firefly[6] to form our PPO dataset. Particularly, we obtained 64,364 and 2,623 for PPO training and testing, respectively[7]. There is no intersection between the training and testing sets. Additionally, we selected 1,704 examples from the SFT test data to create a *forget test set*, enabling us to evaluate the model's ability to retain learned knowledge.

Models We employed BLOOMZ (Muennighoff et al., 2022) as our pre-trained model backbone. More specifically, $\text{BLOOMZ}_{\texttt{7B}}$ was used for reward modeling and $\text{BLOOMZ}_{\texttt{176B}}$ was used for SFT and RLHF training.

4.2 TRAINING SETUPS

We initialized our models using pre-trained checkpoints. The architectural configuration and hyper-parameters were kept consistent with those of the pre-trained models, except that a value head is

[5] https://huggingface.co/datasets/BAAI/COIG

[6] https://huggingface.co/datasets/YeungNLP/firefly-train-1.1M

[7] The PPO training and testing query sets could be shared upon request.

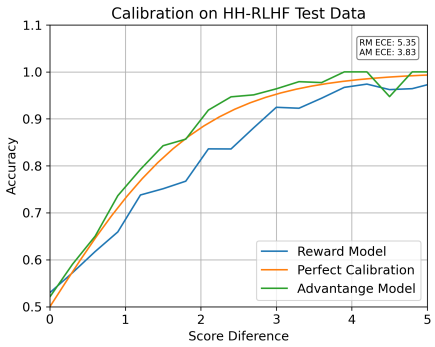
Model	HH-RLHF		Proprietary Data	
	`Accuracy` ↑	`ECE` ↓	`Accuracy` ↑	`ECE` ↓
OpenAssistant Köpf et al. (2023)	69.24	-	-	-
Reward Model	69.25	4.70	74.75	5.35
Advantage Model	69.43	3.48	75.28	3.83

Table 1: Evaluation results on HH-RLHF and our proprietary data. Note that maximizing accuracy is not the exclusive objective in AM optimization. The aim also extends to reducing ECE to improve reliability, whilst sustaining or improving the level of ranking accuracy compared with RM.

added to produce a scalar reward. A learning rate of 5e-6 was employed, coupled with a warm-up strategy covering the initial 10% of training steps and a cosine learning rate schedule decreasing to 10% of the initial learning rate. For the English dataset, a global batch size of 180 was employed, whereas for the Chinese dataset, the batch size was set to 480. The Overfitting issue is observed in general after models are trained for one epoch. As such, we fixed the training epoch as 1 for the all the experiments.For PPO training, a learning rate of 5×10^{-7} and a global batch size of 256 is employed. The actor model is trained for 100 steps for all experiments. The SFT model is trained on the proprietary dataset. We omit these details since these are not the focus of this paper.

4.3 Evaluation

AM Evaluation Results Firstly, we present the overall accuracy and Expected Calibration Error (ECE) for both RM and AM on each dataset. For the English dataset, we additionally compare our method with the publicly available OpenAssistant (Köpf et al., 2023) which utilized DeBERTa (He et al., 2020) for reward modeling. Table 2 lists all the results. We observe that AM achieves slightly higher accuracy but significantly lower ECE on all the datasets. This indicates that AM is capable of maintaining the same level of ranking accuracy while providing reliable and well-calibrated scores. A detailed analysis of calibrations is provided in the following sections. We attribute this phenomenon to the fact that AM is formulated to directly model additional rewards, *i.e.,* advantages, making it more stable and less prone to yield high variances cores. Additionally, the accuracy on the proprietary data is much higher than that on HH-RLHF. We speculate that the trade-off between helpfulness and harmlessness objectives is more pronounced in HH-RLHF, possibly due to the limited presence of harmful examples in our proprietary data.



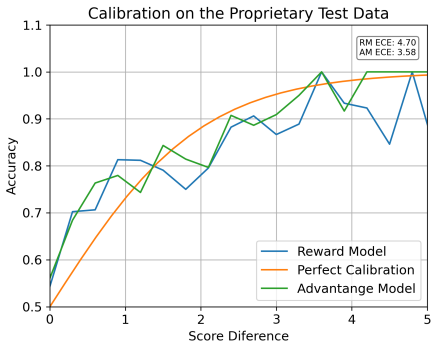
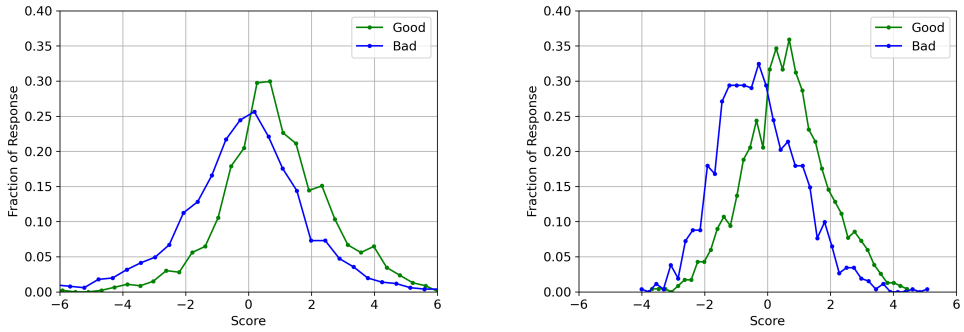


Figure 2: Ranking accuracy is shown as a function of the difference in scores between higher and lower ranked responses. The orange lines indicate the calibrated prediction of accuracy $1/(1+e^{-\Delta})$ in which Δ denotes the score difference. On the left, we show calibration of RM and AM on HH-RLHF data while on the right we show results for our proprietary data. We observe that AM calibration is better than RM's.

Calibrations of AM The reward model score of a response should accurately reflect the probability that humans prefer it. These probabilities must be precise; in other words, the scores should be

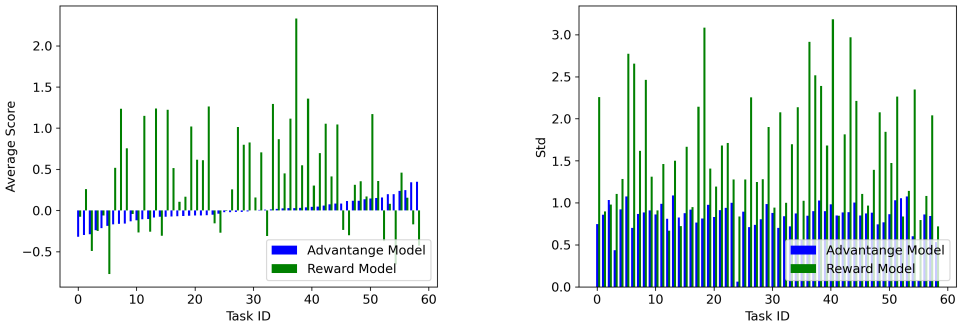


(a) RM score distribution.

(b) AM score distribution.

Figure 3: Distributions of RM and AM scores for pairs of good and bad examples from the proprietary data.

well-calibrated. This is crucial since these scores will serve as reward signals to guide PPO training Bai et al. (2022a). To assess whether our AM is calibrated or not, in Figure 2, we depict the ranking accuracy as a function of score differences assigned to pairs of samples. An orange line representing perfect calibration is also included. Our observations indicate that the AM exhibits significantly lower ECE and is better calibrated than RM on both datasets, whereas RM tends to be overconfident in most cases. We further show the distribution of scores for both good and bad examples in Figure 3. While in general both RM and AM are able to assign higher scores for good examples, AM exhibits a more distinct distribution pattern.



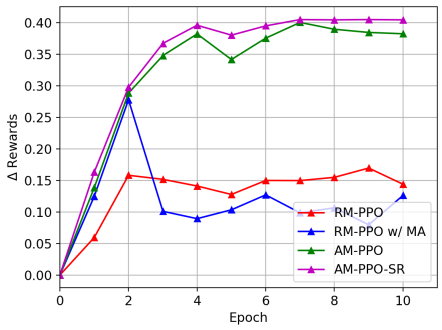
(a) Mean scores of RM and AM for each task.

(b) Std of RM and AM for each task.

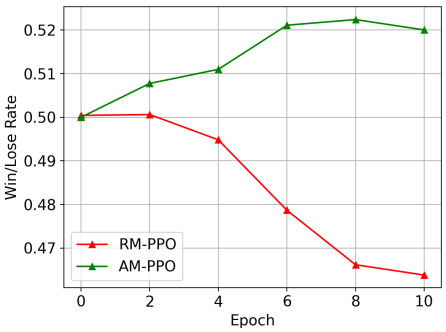
Figure 4: Mean and standard variance for each task categorized by a task spectrum on the in-house data.

Means and variances of AM During PPO training, RLHF exhibits instability, largely owing to unpredictable fluctuations in reward estimation scales. Directly modeling advantage, as our AM does, could potentially alleviate the above issue. To validate AM's efficacy in stabilizing score scales and ranges, we calculated the AM scores for individual examples and analyzed the mean and variance across all the the task spectrum. This analysis is depicted in Figure 4a. We observe markedly different means for each task in the case of RM. Such significant disparities in means can potentially give rise to reward hacking issues (Skalse et al., 2022) and result in repeated failures during PPO training. In addition, Figure 4b illustrates the standard deviations of both AM and RM, with AM consistently operating at a stable scale. These results endorse AM as a strategy designed to normalize reward scores at the individual example level while enhancing ranking accuracy.

PPO training results We conducted a comparative analysis of PPO training with different scoring models in terms of their performance on both main test set and forget test set. The learning curve



(a) Learning curves of various models on delta rewards



(b) Win/Loss rate over SFT model evaluated by GPT-4.

Figure 5: PPO training curves on the Main Test Set with different scoring models. RM-PPO and AM-PPO denote PPO trained with Reward Model and Advantage Model, respectively. AM-PPO-SER additionally equips with Selective Rehearsal.

Model	Main Test Set			Forget Test Set		
	`Win` ↑	`Lose` ↓	Tie	`Win` ↑	`Lose` ↓	Tie
RM-PPO	12.72	12.62	74.66	16.87	29.28	53.84
AM-PPO	14.87	10.38	74.74	9.70	8.44	81.86
AM-PPO-SR	15.78	9.77	74.45	10.30	7.95	81.75

Table 2: Comparison results of different models over the SFT model.

is shown in 5. We observe that AM-PPO outperformed RM-PPO in the main set, achieving higher rewards and a superior win rate over the SFT model. In addition, RM-PPO faces significant reward hacking issues, witnessed by a drop in win rate evaluated by GPT-4, shown in 5b despite a rise in RM scores. Despite utilizing moving average for score normalization, RM-PPO w/ MA encounters instabilities during PPO training. Conversely, AM-PPO exhibits resistance to such problems, maintaining stable GPT-4 outcomes. This emphasizes AM's stability and alignment efficiency over RM. The forget test set result reveal RM-PPO's substantial susceptibility to catastrophic forgetting, portraying a noticeable performance drop. In contrast, AM-PPO is stable, avoiding significant drops and showcasing stability. Incorporating selective rehearsal, the AM-PPO-SR variant demonstrate an uplifted win rate on both sets, underscoring the role of selective rehearsal in alleviating catastrophic forgetting and enhancing model efficacy.

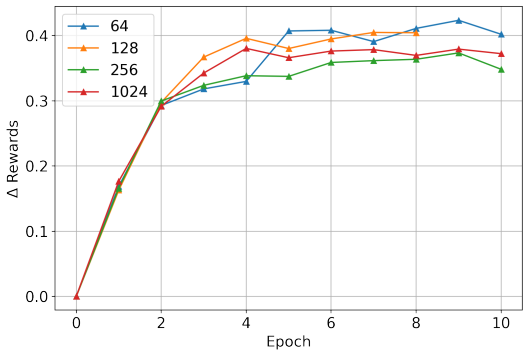


Figure 6: The AM-PPO-SR training curves on the Main Test Set with different number of clustering groups c for selective rehearsal.

Analysis on Selective Rehearsal We also conduct an in-depth examination of the impact of the number of clusters, denoted as c, in the context of selective rehearsal during PPO training. As illustrated in Figure 6, our results reveal a relatively consistent variance of approximately 0.05 points in test-set rewards across various cluster numbers c. While our findings highlight the robustness of the selective rehearsal technique, we recommend conducting a thorough analysis of this aspect when applying selective rehearsal to different datasets, as domain-specific variations can have a notable impact.

5 Related Work

LLM Alignments with Human Preferences. LLMs are typically pre-trained on extensive datasets and can be adapted to a wide variety of downstream tasks. One critical aspect of utilizing LLMs effectively is ensuring their alignment with human preferences, which helps in averting responses that are unsafe, toxic, sexually explicit, biased, or criminal (Leike et al., 2018). A predominant strategy in achieving this is RLHF. This involves training a reward model based on human feedback and utilizing PPO to improve to fine-tuning LLMs (Christiano et al., 2017; Bai et al., 2022a; Glaese et al., 2022; Bai et al., 2022b; Stiennon et al., 2020; Qiu et al., 2022).

Instabilities in RLHF. Despite its success, the RLHF approach is inherently complex and poses significant challenges, thereby encouraging the exploration of simpler methods to align LLMs with human preferences. In this context, Cobbe et al. (2021) introduced the best-of-n sampling, which reinforces LLMs by choosing the responses with the highest reward score from a set of n responses. A similar pathway was pursued by RAFT (Dong et al., 2023), which focuses on selecting high-quality samples to fine-tuning to enhance the model's performance. Moreover, the RRHF strategy (Yuan et al., 2023) evaluates sampled responses from various sources using the logarithm of conditional probabilities. It then aligns these probabilities with human preferences by applying ranking loss, fostering a more refined alignment process. Furthermore, Rafailov et al. (2023) introduced the concept of Direct Preference Optimization (DPO). This approach leverages a relationship between reward functions and optimal policies to address a constrained reward maximization problem through a single stage of policy training. In a similar vein, Preference Ranking Optimization (PRO) (Song et al., 2023) sidesteps the necessity for Reinforcement Learning (RL) training. Instead, it directly aligns LLMs with human preferences using the Bradley-Terry comparison — a method that involves the probability ranking of n responses generated by the LLM, ensuring they are consistent with human preference rankings.

Data Curation for LLM Alignments. Many approaches have been devised to curate high-quality, instruction-following datasets to fine-tune LLMs (Wang et al., 2022; 2023; Taori et al., 2023; Chiang et al., 2023; Peng et al., 2023). For instance, the study by LIMA (Zhou et al., 2023) underscores that even a limited set of carefully curated and high-quality examples can be utilized to fine-tune a strong pre-trained language model, enabling it to deliver competitive results across a diverse array of prompts. Similarly, Wei et al. (2023) introduced a versatile and straightforward data selector designed to autonomously curate a subset from the original fine-tuning dataset, adhering to specific principles for training vision-language models. While these strategies converge on the shared objective of data curation for LLM fine-tuning, our approach is uniquely centered on data curation for PPO training. This strategy diverges fundamentally from others that emphasize the SFT stage, thereby addressing a distinct problem.

6 Conclusion

In this report, we identified and analyzied critical impediments in RLHF training of LLMs, namely reward hacking and catastrophic forgetting. These issues emerge due to the variances in learned reward score distributions and the over-optimization of specific training examples, resulting in instabilities in RLHF training. To alleviate these issues, we introduced the *Advantage Model* and *Selective Rehearsal*—innovative strategies formulated to stabilize the RLHF training process. The Advantage Model aims to maintain balanced reward score distributions across diverse categories and examples, thereby averting complications arising from reward hacking. On the other hand, Selective Rehearsal selectively identifies optimal examples for PPO training, ptimal examples for PPO training, encouraging the retention of crucial knowledge from the SFT stage, and preventing the depreciation of performance over time. Empirical analyses conducted on a range of datasets substantiated the efficacy of our proposed techniques, which not only enhanced stability in RLHF training but also led to improved reward scores and win rates the SFT models.

References

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, et al. A general language assistant as a laboratory

for alignment. *arXiv preprint arXiv:2112.00861*, 2021.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022a.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022b.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, March 2023. URL `https://lmsys.org/blog/2023-03-30-vicuna/`.

Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Hanze Dong, Wei Xiong, Deepanshu Goyal, Rui Pan, Shizhe Diao, Jipeng Zhang, Kashun Shum, and Tong Zhang. Raft: Reward ranked finetuning for generative foundation model alignment. *arXiv preprint arXiv:2304.06767*, 2023.

Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*, 2022.

Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 6894–6910, 2021.

Amelia Glaese, Nat McAleese, Maja Trebacz, John Aslanides, Vlad Firoiu, Timo Ewalds, Maribeth Rauh, Laura Weidinger, Martin Chadwick, Phoebe Thacker, et al. Improving alignment of dialogue agents via targeted human judgements. *arXiv preprint arXiv:2209.14375*, 2022.

Caglar Gulcehre, Tom Le Paine, Srivatsan Srinivasan, Ksenia Konyushkova, Lotte Weerts, Abhishek Sharma, Aditya Siddhant, Alex Ahern, Miaosen Wang, Chenjie Gu, et al. Reinforced self-training (rest) for language modeling. *arXiv preprint arXiv:2308.08998*, 2023.

Kshitij Gupta, Benjamin Thérien, Adam Ibrahim, Mats L Richter, Quentin Anthony, Eugene Belilovsky, Irina Rish, and Timothée Lesort. Continual pre-training of large language models: How to (re) warm your model? *arXiv preprint arXiv:2308.04014*, 2023.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. Deberta: Decoding-enhanced bert with disentangled attention. *arXiv preprint arXiv:2006.03654*, 2020.

Khimya Khetarpal, Matthew Riemer, Irina Rish, and Doina Precup. Towards continual reinforcement learning: A review and perspectives. *Journal of Artificial Intelligence Research*, 75:1401–1476, 2022.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, et al. Openassistant conversations–democratizing large language model alignment. *arXiv preprint arXiv:2304.07327*, 2023.

Jan Leike, David Krueger, Tom Everitt, Miljan Martic, Vishal Maini, and Shane Legg. Scalable agent alignment via reward modeling: a research direction. *arXiv preprint arXiv:1811.07871*, 2018.

Michael McCloskey and Neal J Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation*, volume 24, pp. 109–165. Elsevier, 1989.

Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng-Xin Yong, Hailey Schoelkopf, et al. Crosslingual generalization through multitask finetuning. *arXiv preprint arXiv:2211.01786*, 2022.

R OpenAI. Gpt-4 technical report. *arXiv*, pp. 2303–08774, 2023.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35: 27730–27744, 2022.

Baolin Peng, Chunyuan Li, Pengcheng He, Michel Galley, and Jianfeng Gao. Instruction tuning with gpt-4. *arXiv preprint arXiv:2304.03277*, 2023.

Liang Qiu, Yizhou Zhao, Jinchao Li, Pan Lu, Baolin Peng, Jianfeng Gao, and Song-Chun Zhu. Valuenet: A new dataset for human value driven dialogue system. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 11183–11191, 2022.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward gaming. *Advances in Neural Information Processing Systems*, 35:9460–9471, 2022.

Feifan Song, Bowen Yu, Minghao Li, Haiyang Yu, Fei Huang, Yongbin Li, and Houfeng Wang. Preference ranking optimization for human alignment. *arXiv preprint arXiv:2306.17492*, 2023.

Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. Learning to summarize with human feedback. *Advances in Neural Information Processing Systems*, 33:3008–3021, 2020.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. `https://github.com/tatsu-lab/stanford_alpaca`, 2023.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language model with self generated instructions. *arXiv preprint arXiv:2212.10560*, 2022.

Yizhong Wang, Hamish Ivison, Pradeep Dasigi, Jack Hessel, Tushar Khot, Khyathi Raghavi Chandu, David Wadden, Kelsey MacMillan, Noah A Smith, Iz Beltagy, et al. How far can camels go? exploring the state of instruction tuning on open resources. *arXiv preprint arXiv:2306.04751*, 2023.

Lai Wei, Zihao Jiang, Weiran Huang, and Lichao Sun. Instructiongpt-4: A 200-instruction paradigm for fine-tuning minigpt-4. *arXiv preprint arXiv:2308.12067*, 2023.

Zheng Yuan, Hongyi Yuan, Chuanqi Tan, Wei Wang, Songfang Huang, and Fei Huang. Rrhf: Rank responses to align language models with human feedback without tears. *arXiv preprint arXiv:2304.05302*, 2023.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. Lima: Less is more for alignment. *arXiv preprint arXiv:2305.11206*, 2023.